

Geologically Recent Formation of Some Tesserae on Venus by Plains Deformation



Key Points:

- Some highland tesserae on Venus show deformational evidence for having grown through the incorporation of adjacent plains
- These shortening systems resemble fold-and-thrust belts on Earth and may have formed in a similar manner
- Some of this deformation involves the stratigraphically youngest local plains materials, indicating geologically recent tectonic activity

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Abstract Tessera is a pervasively deformed terrain type on Venus generally interpreted as the oldest preserved material on the surface. Large expanses of this terrain type are typically elevated and have been hypothesized to be felsic, perhaps analogous to Earth's continents and even to date from an era with a more Earth-like climate earlier in Venus' history. However, the compositions, modes of emplacement, and timing of deformation of this terrain type remain poorly understood. Here we propose, on the basis of detailed structural mapping, that volcanic plains along portions of the boundaries of the two largest highlands on Venus have been tectonically deformed so as to acquire the morphological characteristics of adjacent tessera units, thus increasing the areal extent of tessera globally. Some of the incorporated plains units are among the stratigraphically youngest on the planet. Our interpretation suggests that the formation of tessera terrain has continued into the geologically recent past on Venus, and that at least some portion of the crust underlying tessera has not required the production of large volumes of felsic rocks.

Plain Language Summary The surfaces of most highlands on Venus consist of highly deformed terrain known as “tessera.” The nature, formation, and age of tessera remain unknown. On the basis of a combination of structural mapping from radar imagery as well as analytical modeling of stresses in the crust of Venus, we suggest that, at least in places, these highlands have grown through the deformation and incorporation of younger, adjacent volcanic plains. If so, then portions of the tesserae are younger than previously thought, at least some volume of rocks with this terrain type were originally volcanic, and some of this incorporation occurred in geologically recent time.

1. Introduction

The term “tessera” (pl. “tesserae”) describes a major terrain type on the surface of Venus characterized by tectonic complexity and high radar backscatter (Barsukov et al., 1986; Bindschadler et al., 1990) (Figure 1, panel a) and occupying about 7% of the planet's surface area (Ivanov & Head, 2011) (Figure 1, panel b). Surface units mapped as tesserae feature at least two and frequently more sets of cross-cutting shortening and/or extensional tectonic structures indicative of multiple episodes of deformation. Most larger expanses of this terrain type are topographically elevated, and the surfaces of much of the planet's two major highlands, Aphrodite and Ishtar Terrae (Figure 1, panel b), are composed of heavily folded tesserae (Basilevsky & Head, 1998; Burke et al., 1984; Ivanov & Basilevsky, 1993).

Tesserae are generally interpreted as marking the oldest preserved crust on Venus (Basilevsky & Head, 1998; Hansen & López, 2010; Ivanov & Head, 2011), largely on the basis of their common superposition by younger and more widespread radar-smooth plains that are generally regarded as volcanic in origin (e.g., Ivanov & Head, 2011). Tesserae also have a greater areal density of large impact craters (diameter $D \geq 16$ km) than the plains (Ivanov & Basilevsky, 1993). The recognition of many relatively small exposures of tessera far from larger expanses of this terrain type led to the suggestion that tessera underlies large portions of the younger plains (Basilevsky & Head, 1998; Ivanov & Head, 1996).

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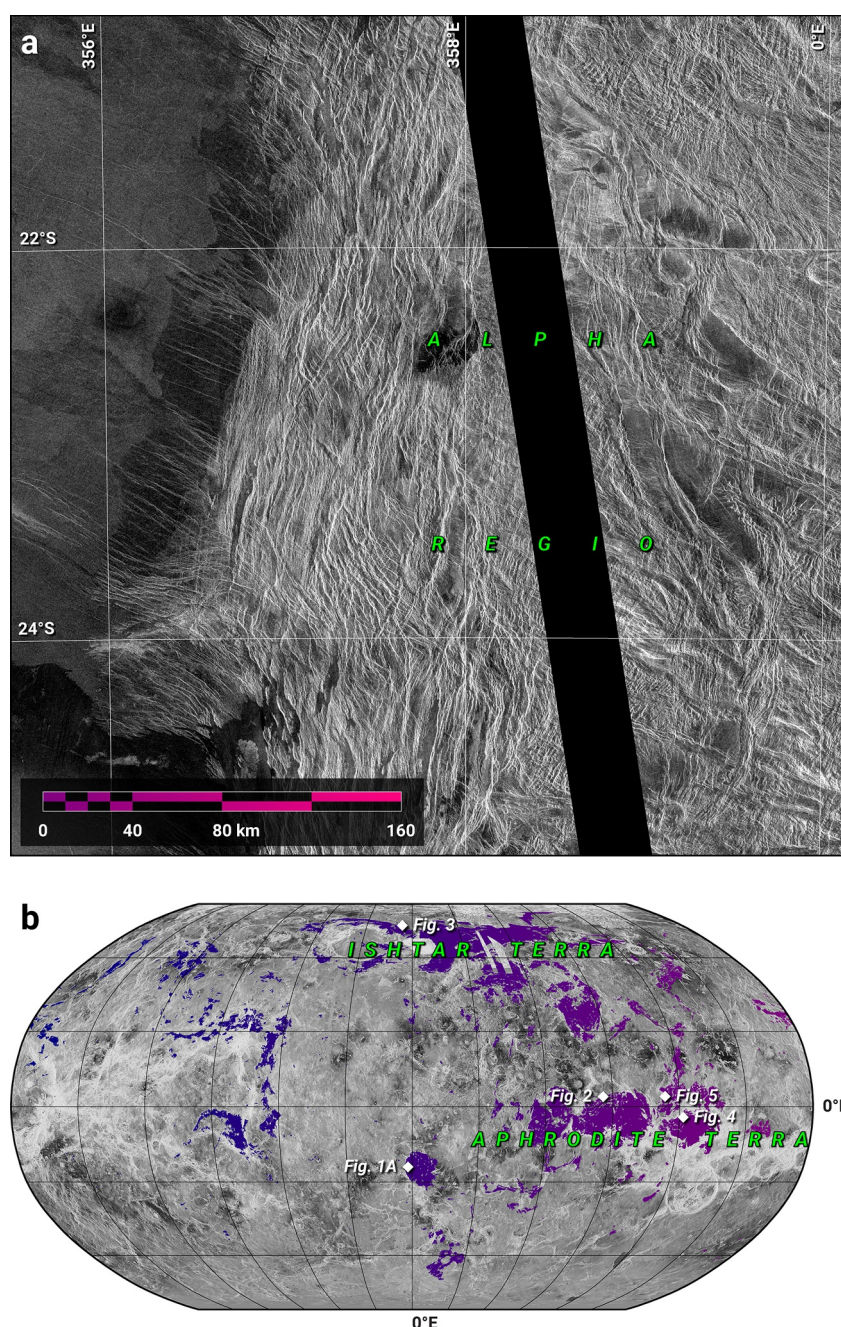


Figure 1. Tesserae on Venus. (a) Magellan synthetic-aperture radar image (S-band, 12.6-cm wavelength) mosaic of a portion of the western margin of Alpha Regio at low southern latitudes on Venus. This tessera exposure, characteristic of tessera units on Venus generally, is heavily deformed by both shortening and cross-cutting extensional structures. The radar look direction is to the east, and the image is in azimuthal equidistant projection centered at 23°S, 358°E; the black area is a gore (data gap) in the Magellan mosaic. (b) A global map of tessera units (in purple, after Ivanov and Head (2011)) atop an east-look global Magellan radar image mosaic in Robinson projection centered at 0.0°N, 0.0°E. The two major highlands on Venus, Aphrodite and Ishtar Terrae, are labeled, as are the locations of panel (a) and Figures 2–5.

However, “tessera” is a descriptor of surface morphology only, and information regarding the rocks that comprise this terrain type is limited. Further, tessera formation was not necessarily everywhere simultaneous (e.g., Guest & Stofan, 1999; Hansen, 2000), and the tesserae preserved today need not have originated from the same rock type, been emplaced in the same way, or been deformed by the same process(es). Different tessera exposures, and even

different regions within a single tessera, may well have different mineralogies and, by implication, different precursor lithologies (Brossier & Gilmore, 2021; Gilmore & Head, 2018).

On the basis of near-infrared emissivity, structural analysis, and gravity anomalies consistent with thicker-than-average crust, it has been suggested that highland tesserae may be felsic and constitute the Venus counterparts to continents on Earth (Hashimoto et al., 2008; Romeo & Capote, 2011; Romeo & Turcotte, 2008). For instance, near-infrared emissivity data from the European Space Agency's Venus Express spacecraft have been interpreted as indicating that the Alpha Regio tessera (Figure 1, panel a) has a lower ferrous iron content and thus may be more silica rich than the surrounding volcanic plains (Gilmore et al., 2015). The formation of large volumes of evolved (i.e., felsic) crust on Earth is thought to have required substantial amounts of surface or near-surface water and a mechanism for recycling that water into the mantle and forming silica-rich magmas by partial melting (Campbell & Taylor, 1983). If the surfaces of highland tesserae are felsic, their formation could reflect a period in the past when Venus had a more Earth-like climate (Hashimoto et al., 2008; Romeo & Capote, 2011; Way et al., 2016), one that might have persisted until as recently as 1 Ga (Way & Del Genio, 2020), together with some mechanism for the recycling of lithosphere and water into the mantle. However, whether the full thickness of highland crust can consist of silica-rich rocks can be questioned on the basis of rheological limits to long-wavelength topographic relief (Nimmo & Mackwell, 2023).

Here, on the basis of detailed structural mapping and analysis from orbital radar imagery acquired by NASA's Magellan spacecraft in the 1990s, we offer a structural interpretation for several regions on Venus where lowland volcanic plains seem to show progressively greater shortening deformation with proximity to the margins of higher-standing tesserae, such that the structures formed in the plains gradually become indistinguishable from the dominant tectonic fabric of the tesserae themselves. Some of this deformation of surface rocks to acquire the morphological characteristics of tesserae may have been geologically recent, that is, younger than the times of emplacement of the stratigraphically youngest local volcanic plains. We therefore hypothesize that, in a manner broadly similar to how continents on Earth grew outward by the addition of other crustal material, the volume and areal extent of some tesserae on Venus increased through the deformation and incorporation of adjacent plains along portions of their margins.

2. Methods

2.1. Structural Mapping

To identify candidate sites of tessera formation on Venus we made use of Magellan synthetic-aperture radar (SAR) east- and west-looking global radar mosaics, which had been constructed in turn from Full-resolution Basic Image Data Record radar images. These mosaics have a stated spatial resolution of 75 m per pixel (m/px), but an effective spatial resolution of about 100 m/px. We surveyed these sites at a view scale of 1:1,000,000 and prepared detailed structural maps of selected smaller areas within each site at a larger view scale of 1:400,000. All mapping was performed in the Esri ArcGIS 10.8 environment.

We established three morphological and physiographic criteria that should be satisfied to classify a given portion of the Venus surface as a candidate site of tessera formation. (a) The presence of radar-smooth plains units situated proximal to ridges. (Radar-smooth materials are those with relatively low radar backscatter, in contrast to the majority of materials that comprise tessera exposures.) There has long been a basis for interpreting ridges on Venus as folds (e.g., Solomon et al., 1992), and shortening structures are widely recognized within tesserae (e.g., Byrne, Ghail, Şengör, et al., 2021; Cofrade et al., 2019; Romeo & Capote, 2011). We paid particular attention to those ridges that appear to postdate stratigraphically young local plains on the basis of previous mapping studies, specifically the deformation of the upper and lower "regional plains" units of Ivanov and Head (2011) (e.g., their Figures 13 and 14). (b) Locations of those folds adjacent to tessera units. (c) A similarity of the approximate trends of these folds and the dominant shortening tectonic fabric within the adjacent tessera.

We interpreted and mapped structural landforms at these candidate sites of recent tessera formation following the approaches we implemented for earlier mapping studies of the geometry and kinematics of tectonic structures on Earth and other planetary bodies, including Venus (e.g., Byrne et al., 2014; Crane & Byrne, 2025; Klimczak et al., 2019; Şengör et al., 2018; Tuckwell & Ghail, 2003). Specifically, we interpreted ridges as anticlinal folds on the basis of a progression in radar backscatter values from low to high that is indicative of a rounded, positive-relief morphology in cross section, an interpretation well established in the study of radar images of the Venus

surface (e.g., Romeo & Capote, 2011). We defined the spatial extents of folds on the basis of contacts either with adjacent smooth plains units or with other folds, and we inferred the down-dip directions of fold limbs and mapped their hinge lines. Extensional structures were identified as relatively straight features showing consistent (bright or dark) radar backscatter along their lengths; we interpreted these features as normal fault scarps. Transpressional and transtensional structures were recognized on the basis of their sigmoidal planforms or trends that make a high angle to the primary structures within which they are associated, together with their inferred lateral kinematics. Fold margins, fold axes, and extensional structures were recorded as segmented polylines in ArcMap. All dimensions given in the text were acquired in ArcMap as geodesic length measurements.

To place our analyses in the context of earlier global stratigraphic models for Venus, we compared our structural mapping with the global geological map of Ivanov and Head (2011), the most recent synthesis of the numerous regional-scale maps for Venus prepared with Magellan data and published at a scale of 1:10,000,000. This global map was based on 27 earlier quadrangle maps derived from Magellan data and published between 1995 and 2011 (Ivanov & Head, 2011). We used the unit descriptions from that study; the mapped tessera exposures in Figure 1, panel (b), are also taken from that study.

Radar image data are subject to distortions arising from mapping three-dimensional landforms onto a two-dimensional surface, and those distortions vary with look direction and incidence angle. As a result, a topographically elevated feature will be shifted toward the radar, whereas a feature closer to the mean elevation used for the image projection will be appropriately positioned, unless the image has been corrected with topographic data of a similar spatial scale (the Magellan data cannot be so corrected with available topographic data). Thus, the outlines of folds, inferred fold traces, and other structures with local relief that we have mapped are likely to be laterally mispositioned. Nevertheless, the lateral offsets between tectonic features of the scales we describe in this study arising from these radar distortions are on the order of only a few pixels—we estimate no more than perhaps 500 m (i.e., five pixels). Since neither our mapping (e.g., fold placement) nor structural analyses (e.g., whether folds are upright or verging) are sensitive to positional errors of ~1–2 km, our conclusions are not fundamentally affected by distortions resulting from variations in look direction or incidence angle of Magellan image data.

We also note that fold vergence—that is, the direction of fold asymmetry corresponding to the direction of tectonic transport along an underlying thrust fault—is sensitive to radar layover, especially for folds with axes close to perpendicular to the radar look direction. We worked to minimize this effect since Magellan radar imagery is not corrected for topography, but the direction of fold vergence does not materially impact our inferences or conclusions.

2.2. Stress Modeling

As a basis for comparison with the orientations of tectonic structures in areas of recent tessera formation, we consider models for the state of stress in the upper crust of Venus. Two potentially important sources of deformation are the stresses imparted by interior flow, as constrained by long-wavelength gravity and topography, and the stresses associated with topographic relief and its approximate compensation by variations in crustal thickness (Byrne, Ghail, Şengör, et al., 2021; James et al., 2013). Although interior flow is a major source of stress and deformation in the plains of Venus (Byrne, Ghail, Şengör, et al., 2021), the usual assumption made in the construction of global flow models, that viscosity in the Venus interior varies only with depth, can be questioned for highland tesserae and their boundaries with surrounding plains. Long-wavelength variations in gravity and topography on Venus indicate that there are mass deficits in the mantle beneath both Ovda and Ishtar Terrae (Herrick & Phillips, 1992; James et al., 2013); whether these deficits are thermal or compositional in origin is not known. If, like the stable interiors of continents on Earth, the mass deficits are the result of partial melt extraction (e.g., Carlson et al., 2005), then Ovda and Ishtar Terrae could have deep mantle roots substantially higher in viscosity than typical mantle over the same depth range, a condition that would markedly affect mantle flow beneath and adjacent to these highlands.

More straightforward is the modeling of crustal stress associated with variations in topography and crustal thickness. As quantified by James et al. (2013), the topography of Venus can be characterized as a superposition of dynamic topography (H_d) supported by mantle convective stresses and the portion of the topography that is isostatically supported by crustal thickness variations (H_c). Even in an effectively static state, the presence of any crustal thickness variation requires the existence of horizontal loads in the crust. Following the thin-shell

methodology and terminology of Banerdt (1986), a horizontal—or “tangential”—load vector $\mathbf{q}_T$ in a planet's lithosphere may be expressed as the gradient of a scalar potential function, Ω , that is,

$$\mathbf{q}_T = -\nabla\Omega. \quad (1)$$

This formulation ignores toroidal loads, which can be substantial when the mechanical properties of the lithosphere are laterally heterogeneous (O'Connell et al., 1991). We can use thin-shell theory to estimate the stresses in a lithosphere from any potential function Ω . Here, we calculate the stress arising from H_c but omit any stress associated with dynamic topography H_d . For this specific scenario of crustal isostasy, the potential function is approximated as

$$\Omega = \frac{\nu}{1-\nu} g \rho_c \frac{T_c}{R} H_c, \quad (2)$$

where ν is Poisson's ratio, g is gravitational acceleration, ρ_c is the density of the crust, T_c is the thickness of the crust, and R is the radius of the planet. From this potential function, the stress tensor associated with H_c can be calculated across the planet as described by Banerdt (1986). In the study by James et al. (2013), for instance, H_c and H_d were determined to spherical harmonic degree and order 120 from Magellan gravity and topography.

The analysis here takes a slightly different approach: we represent H_c to higher resolution by subtracting the previously inferred H_d from the full degree-and-order-719 topography model of Wieczorek (2015). To emphasize lateral variations of stresses in the vicinity of the highland–plains boundaries that are the focus of this study, we have added an isotropic horizontal compressive stress of 20 MPa, which yields approximately neutral stress states in the centers of Ovda Regio and Lakshmi Planum. Such an isotropic horizontal stress, which does not affect lateral variations in the stress field from crustal isostasy, can arise from unmodeled sources of stress such as temporal changes in the planet's surface temperature or interior volume.

Although the methodology described by Banerdt (1986) is based on the assumption that the resulting stresses are distributed evenly with depth through the lithosphere, stresses tend to be attenuated at the top and bottom of a lithosphere through brittle failure and plastic deformation, respectively (e.g., Burov & Watts, 2006). Following the methodology of our earlier study (Byrne, Ghail, Şengör, et al., 2021), we can define a non-dimensional weighting function, $\tilde{\mu}$, that is proportional to the local viscosity, μ . If radial shear stresses are negligible—as would be expected in a mechanically thin elastic lithosphere with a free surface—then the non-lithostatic stress components become

$$\sigma_{\theta\theta} = \frac{2\tilde{\mu}(r)}{(1-\nu)(-l(l+1)+2)} \left[\frac{\partial^2 \Omega}{\partial \theta^2} + \nu \left(\frac{\partial^2 \Omega}{\partial \phi^2} \csc^2 \theta + \frac{\partial \Omega}{\partial \theta} \cot \theta \right) \right], \quad (3)$$

$$\sigma_{\phi\phi} = \frac{2\tilde{\mu}(r)}{(1-\nu)(-l(l+1)+2)} \left[\left(\frac{\partial^2 \Omega}{\partial \phi^2} \csc^2 \theta + \frac{\partial \Omega}{\partial \theta} \cot \theta \right) + \nu \frac{\partial^2 \Omega}{\partial \theta^2} \right], \text{ and} \quad (4)$$

$$\sigma_{\theta\phi} = \frac{2\tilde{\mu}(r)}{(-l(l+1)+2)} \frac{\partial}{\partial \theta} \left(\frac{\partial \Omega}{\partial \phi} \csc \theta \right), \quad (5)$$

where r , θ , and ϕ are spherical coordinates and l is the spherical harmonic degree. We further redistribute stresses in the lithosphere by introducing near-surface brittle failure. For σ_1 as the most compressive principal stress and a vertical lithostatic stress, the maximum stress allowed prior to frictional sliding at a depth z is (Turcotte & Schubert, 2014)

$$\sigma_1 \leq \frac{2f\rho_c g z}{\sqrt{1+f^2}-f},$$

where f is the coefficient of friction. With this constraint, we reduce stresses in the uppermost lithosphere and amplify stresses in the remainder to maintain a constant horizontal load vector $\mathbf{q}_T$. We use this final depth distribution to calculate the non-lithostatic stress tensors at crustal depths in a model with a crustal density of 2,800 kg/

m^3 , a mantle density of $3,300 \text{ kg/m}^3$, a Poisson's ratio of 0.25, a coefficient of friction of 0.6, and a surface gravitational acceleration of 8.9 m/s^2 . Such a crustal density is appropriate for basaltic or gabbroic compositions; different choices of density would positively correlate with crustal root relief (Broquet et al., 2025). Crustal stress magnitudes vary in proportion to crustal density (Equation 2), but principal stress directions are only weakly dependent on this parameter. The viscosity profile is that of Model 1 of Byrne, Ghail, Şengör, et al. (2021).

3. Results

3.1. Structural Mapping

On the basis of the criteria for identifying areas of potential tessera formation given above, we have identified four candidate sites. Given that Venus has a surface area greater than the land area of Earth by a factor of three, however, we should emphasize that ours was not an exhaustive survey, and there are likely other regions on Venus where plains have been shortened so as to acquire the morphological and radar backscatter characteristics of nearby tesserae.

The first site where we propose that tessera formation has taken place through deformation of adjacent plains is at the Nayunuwi Montes along the northern margin of Ovda Regio, in the western portion of Aphrodite Terra (Figure 2); this margin corresponds to a subtype of the “Type II” tessera boundary described by Ivanov and Head (1996). There, the southern portion of the abutting radar-smooth plains has been deformed by ~kilometer-wide ridges, which we interpret as anticlinal folds. This interpretation accords with earlier analyses of tectonic structures on Venus in which, for example, wrinkle ridges are mapped as deforming (and thus post-dating) radar-smooth plains (cf. Ivanov and Head (2011), and references therein). These anticlines trend approximately northeast–southwest and broaden, over a distance of about 20 km, to widths of a few kilometers toward the southeast (Figure 2, panel b). Some of these wider folds show rounded termination points, consistent with doubly plunging or periclinal folding (Figure 2, panel c).

The relative timing of folding at this site is notable. In at least one location the folds have been embayed by radar-smooth plains (Figure 2, panel c, blue arrows) on the basis of there being clear margins to the ridges that are consistent morphologically with pronounced changes in slope as the younger plains units abut the fore- and backlimbs of the fold. In other places, however, the folds display no such margins—at least at the resolution of the Magellan radar imagery—suggesting that rather than being the narrow crests of larger folds that are now largely buried, they deform and thus postdate these plains (Figure 2, panel b, cyan arrows).

These plains correspond to the two stratigraphically youngest plains units (“lobate plains” and “smooth plains”) on the global geological map of Venus of Ivanov and Head (2011). The wider folds were mapped by Ivanov and Head (2011) as part of a third plains unit (“ridged plains”) that these authors distinguished from the first two and interpreted to be stratigraphically older.

Farther south, the rocks comprising the mapped tessera of Ovda Regio proper (Ivanov & Head, 2011) show a dominant tectonic fabric that parallels the general northeast–southwest trend of the plains folds; that fabric is manifest as a set of broad, continuous rises that we again interpret as folds, though at around 20 km in width they are much broader than their counterparts in the plains (Figure 2, panel d). This dimension is shared by at least one of the folds in the “ridged plains” to the north (Figure 2, panel e). The tessera terrain also hosts narrower folds of the same overall northeast–southwest trend (Figure 2, panel f). The region of deformation from undeformed plains to tessera here is about 60–70 km in width.

A secondary fabric of parallel normal faults and graben trends approximately orthogonal to, and crosscuts, the broad folds (e.g., Figure 2, panel g). Similar extensional landforms elsewhere in Ovda Regio (Gilmore & Head, 2018) and Ishtar Terra (Smrekar & Solomon, 1992) have been attributed to gravitational spreading of elevated terrain following the formation of highland tesserae. Alternatively, because these structures are approximately orthogonal to the major fold axes, they are kinematically compatible with folding and shortening along the tessera margins.

We interpret these observations collectively to suggest that the radar-smooth plains at this location have been progressively deformed to the southeast by both shortening and extension, such that the plains take on the morphological and radar characteristics of tessera without making any distinct structural boundary with older terrain. Further, the dominant direction of shortening appears to have remained generally consistent between the

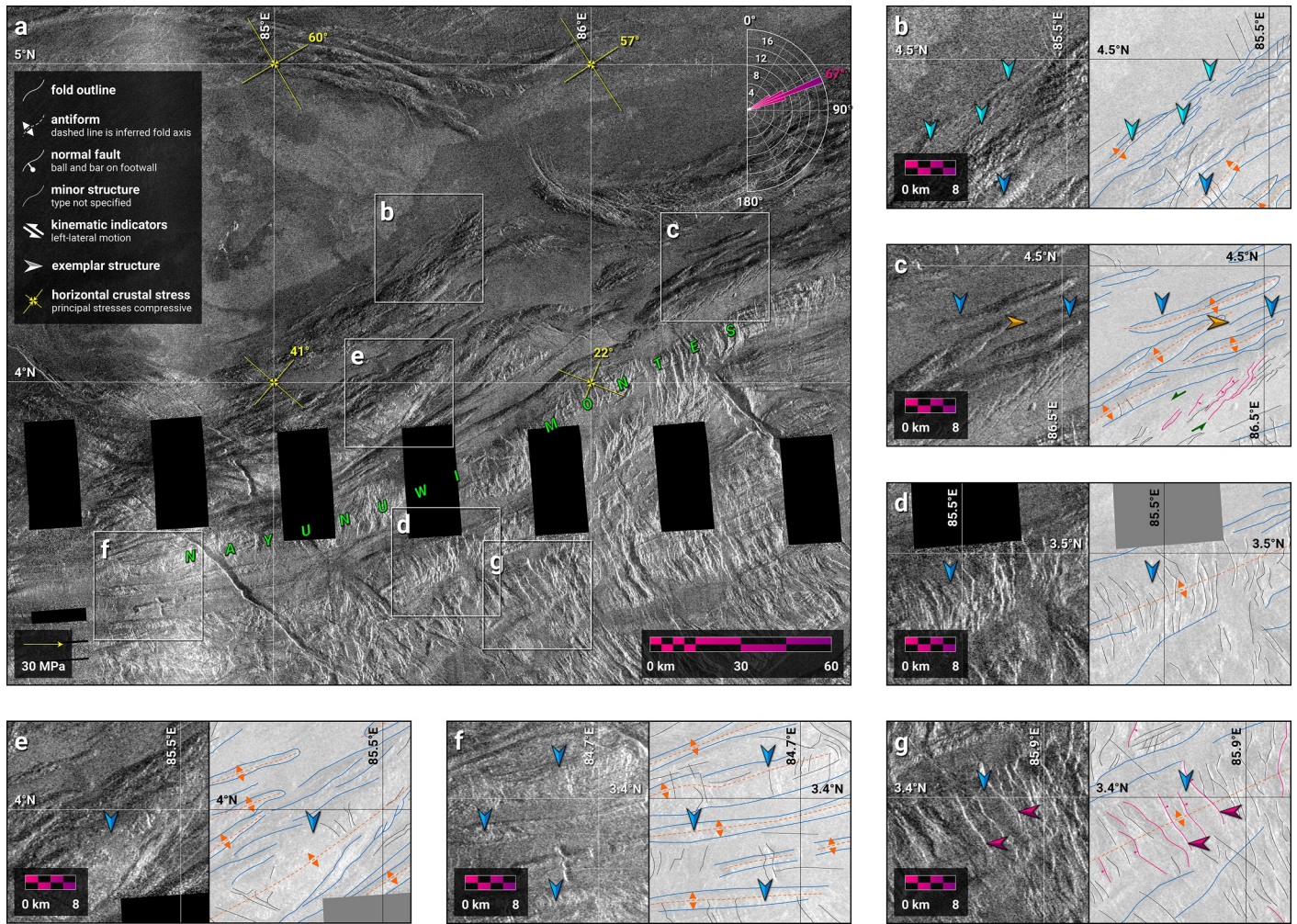


Figure 2. Tessera formation of plains material along the Nayunuwi Montes on the northern margin of Ovda Regio. (a) Magellan radar image mosaic of the region, with Nayunuwi Montes labeled, the outlines of panels (b–g) shown, and explanations for symbols appearing in this and other panels given in a legend. (b) Radar image (left) and structural sketch (right) of folds ~1 km wide (examples are shown with cyan arrows) that we interpret to have deformed the stratigraphically youngest radar-smooth plains here and which broaden to the southeast (blue arrow). (c) Broader folds a few kilometers wide and several tens of kilometers long, some with rounded termination points, situated between the smooth plains units and the tessera comprising Ovda Regio and showing a dominant northeast–southwest trend. In at least one location, radar-smooth plains appear to have embayed these folds (gold arrow). (d) Broad, ~20-km-wide folds in the tessera that share the same general trend as the narrower folds to the north. (e) A similarly sized ~20-km-wide fold, again trending northeast–southwest, between the plains and the tessera. (f) Narrow folds comparable to and with the same prevailing trend as those in the radar-smooth plains within the tessera unit itself. (g) A secondary fabric of extensional structures (magenta arrows) trending orthogonal to the narrow and broad folds within the tessera. In the sketch maps in this and subsequent figures, shortening structures are outlined in blue, interpreted fold axes are shown with dashed orange lines (with orange triangles denoting down-limb directions), and prominent extensional structures are in magenta (with ball-and-bar symbols shown on example footwalls). Minor and/or poorly expressed fractures of various types are shown as thin, black lines. The crossed yellow arrows in panel (a) show the directions and relative magnitudes of modeled horizontal principal stresses at 5 km depth in the crust from variations in topography and crustal thickness, plotted every degree in latitude and longitude (see lower left for scale). The yellow numbers correspond to the predicted trend (measured clockwise from north) of shortening structures formed under these stresses. The average orientations of 50 representative structures at this site are shown in the half-polar plot at the upper right in panel (a). The radar look direction is to the west in all panels, which are in azimuthal equidistant projections centered at (a) 4.0°N, 86.5°E; (b) 4.4°N, 85.5°E; (c) 4.5°N, 86.6°E; (d) 3.3°N, 85.5°E; (e) 3.9°N, 85.3°E; (f) 3.4°N, 84.8°E; and (g) 3.3°N, 85.7°E. The black areas are gores in the Magellan mosaic.

times of deformation of the plains and the immediately adjoining tessera. We therefore propose that there is a general increase in shortening strain from northwest to southeast across the plains of this margin, with the southern limit of the “lobate plains” and the “smooth plains” here modestly shortened and the “ridged plains” more deformed but still less so than the tesserae themselves.

The tectonic pattern along the plains adjacent to Nayunuwi Montes is also present along Uorsar Rupes, on the boundary between Snegurochka Planitia and Ishtar Terra, the other major highland on Venus (Figure 3). At this second site, radar-smooth plains again show narrow folds (Figure 3, panel b) that broaden in width toward and

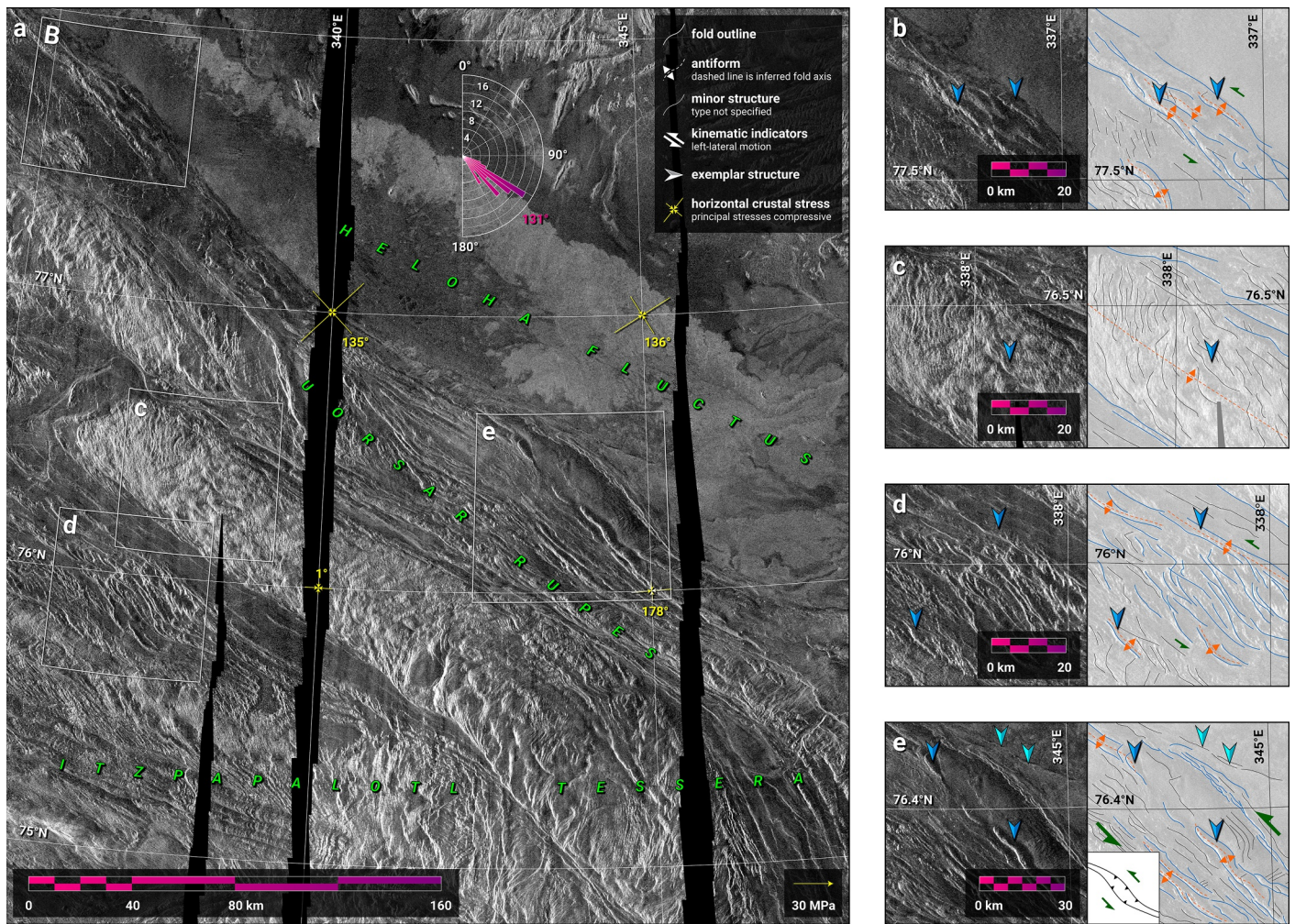


Figure 3. Tessera formation of plains material in Uorsar Rupes, along the Snegurochka Planitia–Ishtar Terra boundary. (a) Magellan radar image mosaic of the region, with prominent physiographic features labeled, the outlines of panels (b–e) shown, and explanations for symbols appearing in this and other panels given in a legend. (b) The southwestern extent of the plains unit here features a set of narrow, parallel folds that trend to the northwest. (c) In the tessera to the south, broader folds share the same northwest–southeast trend. (d) Narrower, northwest-trending folds are also present within the tessera. (e) Sigmoidal folds at the plains–tessera boundary suggest left-lateral transpression in addition to orthogonal shortening (green kinematic arrows in inset); there is also minor top-to-the-left transpression in the areas shown in panels (b and d). Narrow (~1 km-wide) folds deform the stratigraphically youngest radar-smooth plains here as well (cyan arrows). The yellow arrows and numbers and half-polar plot in panel (a) are as for Figure 2. The radar look direction is to the east in all images, which are in azimuthal equidistant projections centered at (a) 76.0°N, 343.0°E; (b) 77.7°N, 336.3°E; (c) 76.3°N, 338.4°E; (d) 75.9°N, 337.6°E; and (e) 76.2°N, 343.2°E. The black areas are gores.

within the tessera (Figure 3, panel c). Narrower folds are present within the tessera (Figure 3, panel d), and all shortening landforms share a broadly northwest–southeast trend. Notably, there are narrow folds along this tessera margin with the same general trend that appear to postdate the radar-smooth plains units (Figure 3, panel e), which again are mapped as the stratigraphically youngest in the region (Ivanov & Head, 2011). The entire zone of deformation from the plains to the tessera interior is around 80 km wide.

This pattern of deformation is also seen in northern Thetis Regio (Figure 4), along the northern margin of Aphrodite Terra and 3,800 km to the east–southeast of the example shown in Figure 2, and in Sogolon Planitia along its boundary with northern Aphrodite Terra (Figure 5), 2,800 km to the east of the site in Figure 2. As with the examples in Figures 2 and 3, we interpret the volcanic plains at these additional sites to be less deformed than, but to host shortening structures that are kinematically similar to, those that comprise the dominant tectonic fabrics of the adjacent tesserae. Each of these sites also features narrow folds that in places seem to postdate and deform the stratigraphically youngest local volcanic plains (Figure 4, panel b and Figure 5, panel b). As at Nayunuwi Montes and Snegurochka Planitia, the disrupted plains units at northern Thetis Regio and Sogolon Planitia are among the youngest major volcanic units on the planet (Ivanov & Head, 2011).

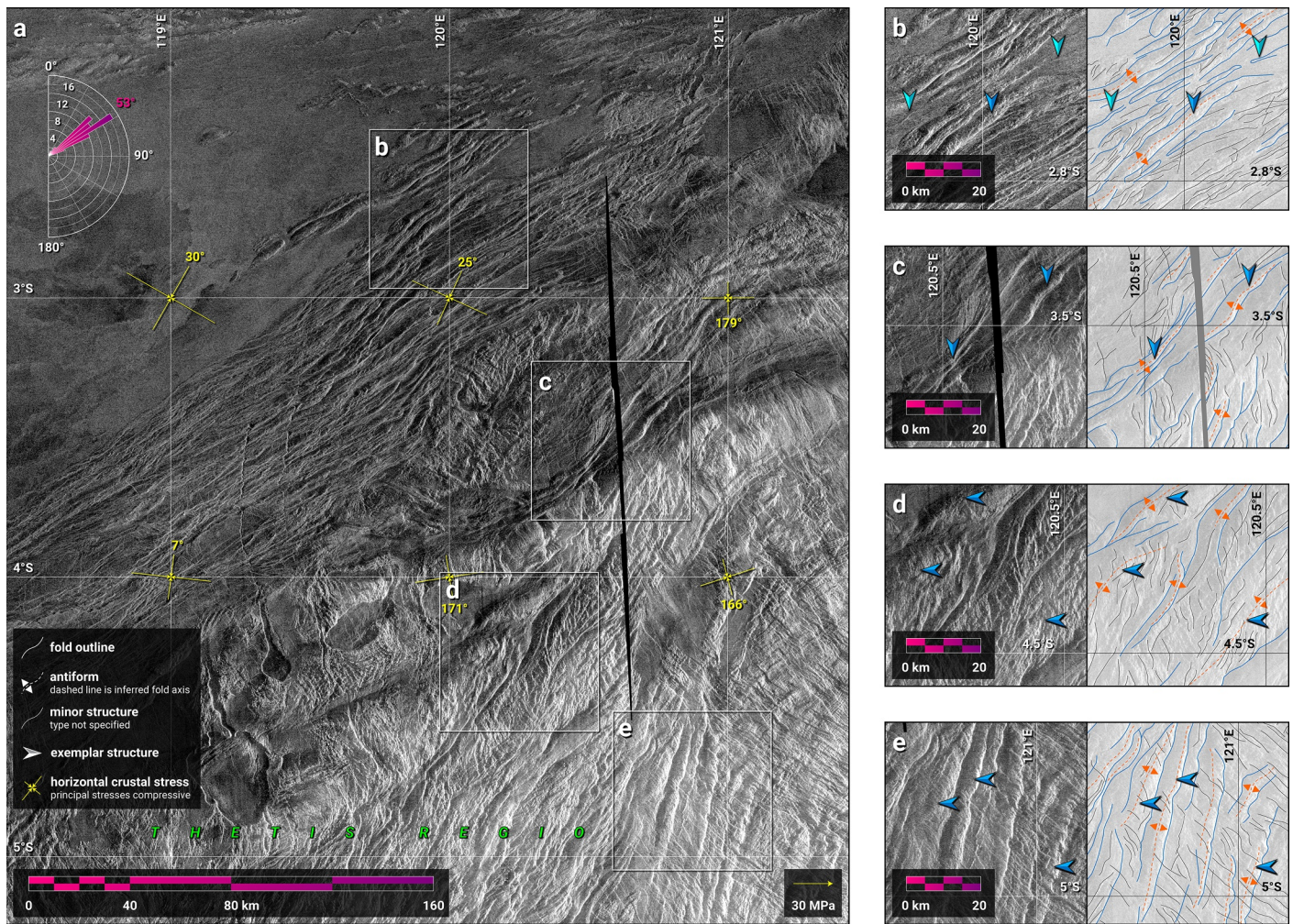


Figure 4. Tessera formation of plains material in northern Thetis Regio. (a) Magellan radar image mosaic of the region, with Thetis Regio labeled, the locations of panels (b–e) shown, and explanations for symbols appearing in this and other panels given in a legend. (b) Narrow, ~1-km-wide folds (cyan arrows) outboard of the tessera with a northeast–southwest trend; these folds deform the stratigraphically youngest radar-smooth plains here and broaden to the southeast (blue arrows). (c) Further examples of narrow folds abutting and having the same overall trend as the broader folds within the tessera unit here. (d) Open, ~10–20-km-wide intra-tessera folds, also trending northeast–southwest, between which are interspersed subparallel, narrower folds. (e) Folds ~10 km across trending north–northeast that constitute the locally dominant tectonic fabric, which can be traced about 100 km northward before curving to the northeast. The yellow arrows and numbers and half-polar plot in panel (a) are as for Figure 2. The radar look direction is to the west in all images, which are in azimuthal equidistant projections centered at (a) 3.6°S, 119.8°E; (b) 2.7°S, 119.8°E; (c) 3.8°S, 120.4°E; (d) 4.3°S, 120.3°E; and (e) 4.7°S, 120.8°E. The black areas are gores.

3.2. Crustal Stress Models

The directions of principal horizontal stresses near the Venus surface from the adopted model of stresses arising from variations in topography and crustal thickness are shown in Figures 2–5. In several locations, the directions of greatest modeled horizontal compression in the areas of recent tessera formation are well aligned with the principal shortening axes of the mapped tectonic structures, that is, the directions orthogonal to the trends of those structures. This alignment is particularly notable outboard of the tessera proper in Figure 2, where deformation of the stratigraphically youngest local plains units is involved (Figure 2, panel a, upper right); along the narrow folds in the stratigraphically youngest local plains at center left in Figure 3; at the margin of the tessera in Figure 4 (just below center); and inboard of the plains–tessera margin in Figure 5 (lower center and upper right).

At other locations, the modeled directions of greatest horizontal compression are not as well aligned with the principal shortening axes of tectonic structures, but some such disagreements are expected given the restriction of the model to long wavelengths because of limits to the resolution of shorter-scale features in current

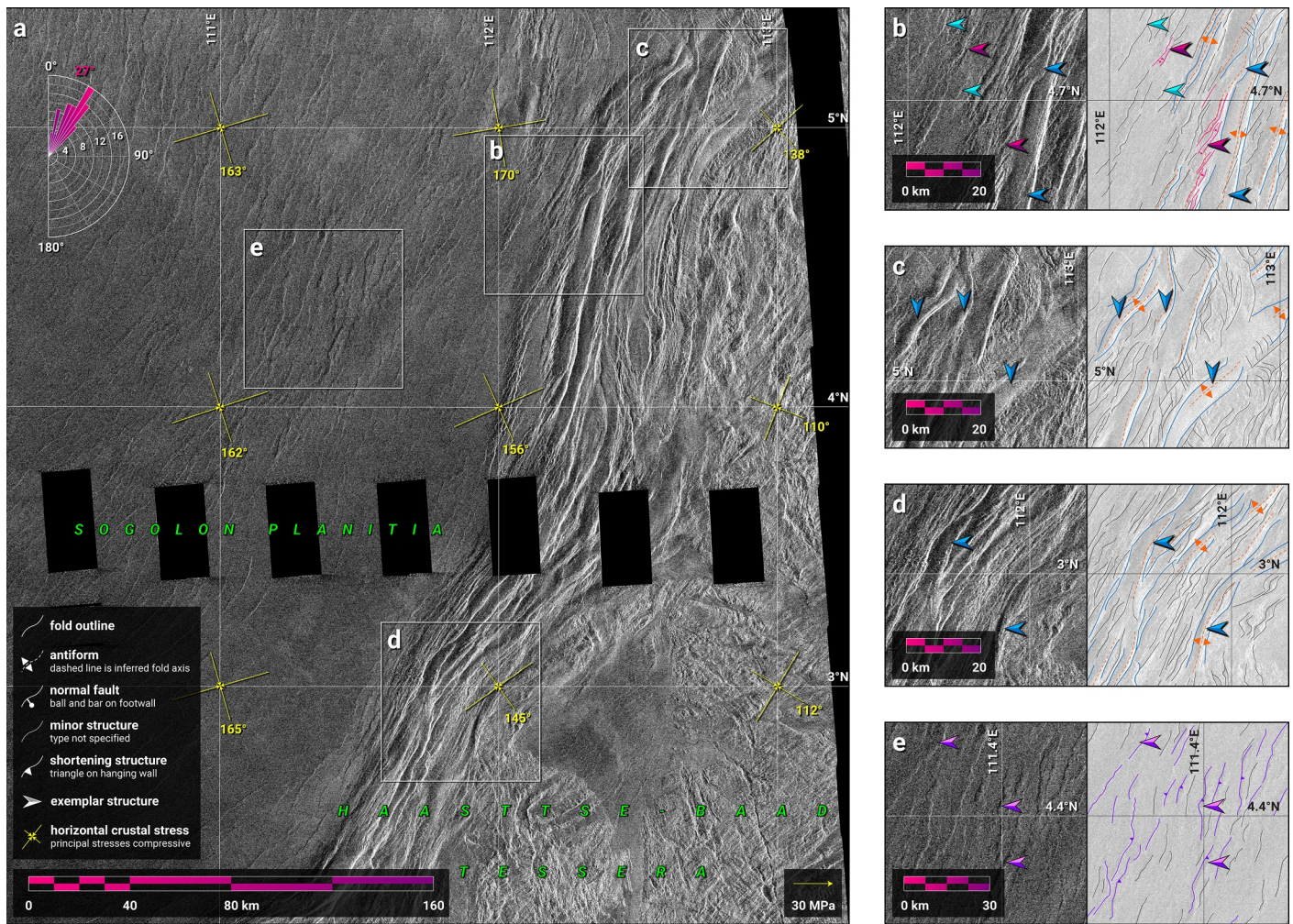


Figure 5. Tessera formation of plains material at the Sogolon Planitia–Aphrodite Terra margin. (a) Magellan radar image mosaic of the region, with Sogolon Planitia and Haastse-baad Tessera (a portion of Aphrodite Terra) labeled, the locations of panels (b–e) shown, and explanations for symbols appearing in this and other panels given in a legend. (b) Folds a few kilometers wide and several tens of kilometers long at the border of the plains and tessera that deform the stratigraphically youngest plains (cyan arrows). (c) Broader, more open folds than in panel (b) at the plains–tessera interface show a change in trend concordant with that of the dominant tessera fabric here. (d) Narrow (~1–2-km-wide) folds within the plains grade into wider folds with proximity to the tessera. (e) Narrow folds in the plains (purple arrows) with the same overall trend as both those at the plains–tessera margin and the dominant tectonic fabric in the tessera. The yellow arrows and numbers and half-polar plot in panel (a) are as for Figure 2. The radar look direction is to the west in all panels, which are in azimuthal equidistant projections centered at (a) 3.5°N, 112.0°E; (b) 4.4°N, 112.5°E; (c) 4.8°N, 112.8°E; (d) 2.7°N, 111.8°E; and (e) 4.5°N, 111.5°E. The black areas are gores.

representations of the topography and gravity field of Venus. Moreover, lateral shearing likely introduces additional complexity to the structures we map, and other processes may contribute to the actual near-surface stress field.

Overall, these findings are consistent with our hypothesis of tessera formation having proceeded outward from the boundaries of highland tesserae by the deformation of adjacent plains over some interval that continued after the time of most recent local plains emplacement.

4. Discussion

4.1. Comparison With Crustal Shortening on Earth

The overall strain pattern we describe here morphologically resembles numerous sites of continental crustal shortening on Earth, such as the Jura Mountains (Figure 6, panel a), the Appalachian Valley and Ridge province (Figure 6, panel b), and the Alice Springs orogeny (Figure 6, panel c) (Buxtorf, 1916; Pfiffner, 2006, 2017;

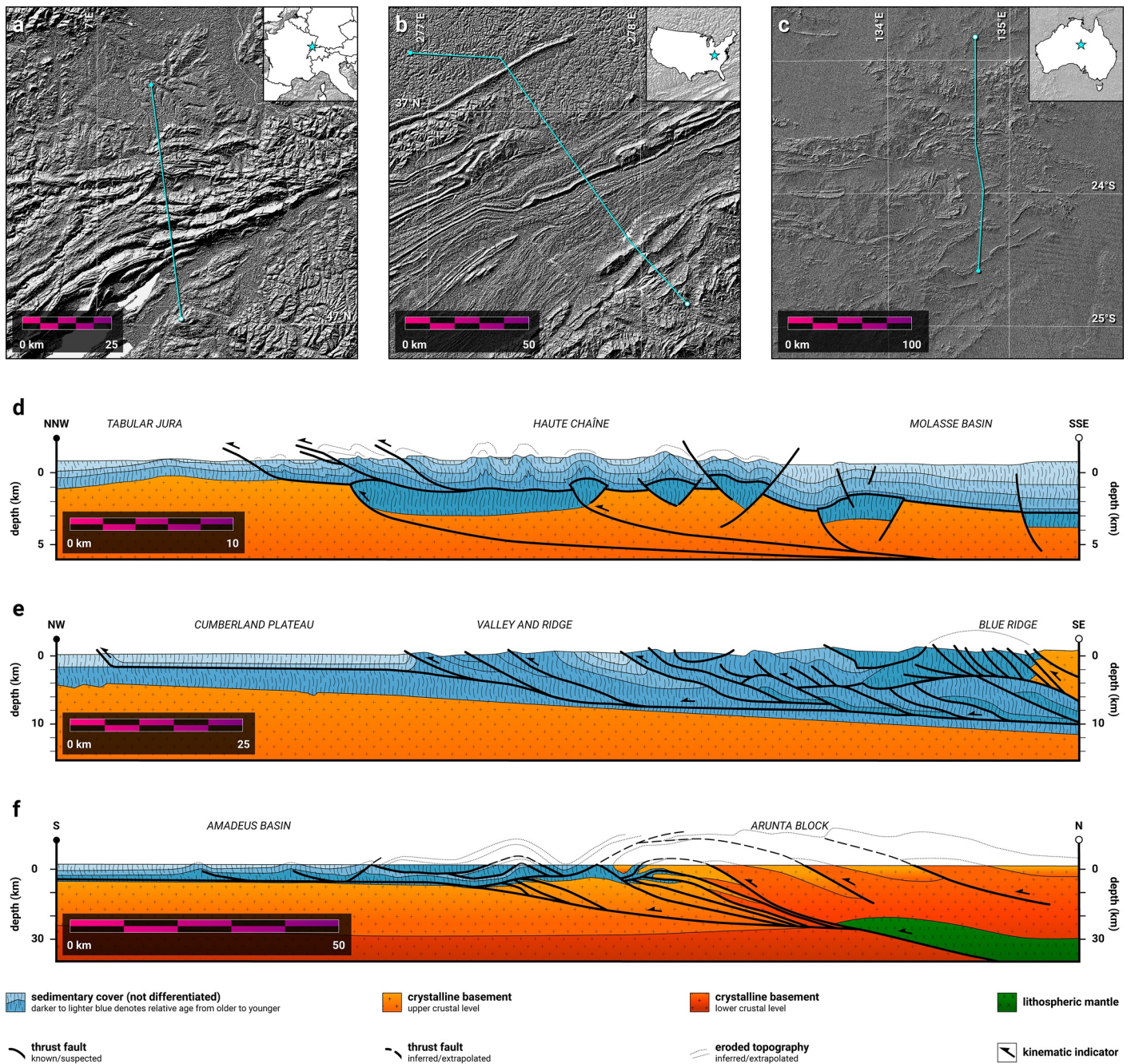


Figure 6. Examples of sites of major continental crustal shortening of different spatial extents on Earth. (a) The Jura Mountains of Switzerland and France. (b) The southern Appalachians in the United States. (c) The Alice Springs orogen in central Australia. In panels (a–c), for comparison with the Magellan synthetic-aperture radar imagery in Figures 1–5, these shortening systems are shown as hillshades derived from the global ASTER (Advanced Spaceborne Thermal Emission and Reflection Radiometer) 30 m/px digital elevation model, with illumination from the northwest and 45° from horizontal. The insets at the upper right in each show the location of each shortening system (marked with a cyan star) in a broad map view. The blue lines in panels (a–c) denote the profiles of the cross sections shown in panels (d–f). (d) Cross section across the Jura, adapted from Pfiffner (2006). (e) Cross section across the southern Appalachians, adapted from Pfiffner (2017), in turn based on Hatcher et al. (1989). (f) Cross section across the Alice Springs orogen, adapted from Pfiffner (2017), in turn based on Teyssier (1985) and Korsch et al. (1998). In panels (d–f), the sedimentary cover is shown in shades of blue with vertical stippling, but is not otherwise differentiated. The crystalline basement is shown in orange; in panel (f), the lower crustal basement is in dark orange, and the lithospheric mantle is in green. Thick black lines denote major thrust faults; where those structures are extrapolated above modern topography, they are dashed. Thin, dotted lines correspond to restored topography (extrapolated from modern fold limbs). Kinematic indicators mark the direction of tectonic transport along select thrust faults. The horizontal:vertical ratios in the cross sections shown in panels (d–f) are 1.0:0.9, 1.0:1.0, and 1.0:0.4, respectively.

Teyssier, 1985; Williams & Hatcher, 1982). These settings are characterized by a combination of folding and thrust faulting, with anticlinal folds often showing asymmetries (i.e., vergence) in the dominant direction of tectonic transport. In some instances, deformation has been mainly restricted to the sedimentary cover (i.e., thin-skinned tectonics) such as in the Jura (Figure 6, panel d), where late Mesozoic and Cenozoic units were shortened atop a décollement (basal detachment) of evaporites (Pfiffner, 2006). A similar structural arrangement is present within the Appalachian Valley and Ridge province (Figure 6, panel e), where shortening along two major décollements deformed late Proterozoic to Phanerozoic platform sediments and clastic deposits (Lammie et al., 2020), forming regularly spaced imbricate thrust systems, but the underlying Grenvillian basement rocks were unaffected (Hatcher et al., 1989; Pfiffner, 2017).

In contrast, both crystalline and sedimentary rocks within the Alice Springs orogeny have been shortened, that is, via thick-skinned tectonics (Figure 6, panel f). The late Proterozoic to early Phanerozoic sediments within the Amadeus Basin were detached from the crystalline basement (Pfiffner, 2017), but imbricate thrust faults penetrated into the lower crust (Teyssier, 1985) and might even have offset the crust–mantle boundary (Korsch et al., 1998). Within the Jura, some deep-seated faulting indicates that shortening of the basement also took place (Pfiffner, 2006). Broader (albeit now largely eroded) folds are associated with these larger, deeper faults (Figure 6, panel d), as is the case above the large thrusts in the Alice Springs system (Figure 6, panel f).

We have little insight into the subsurface structure of the plains–tessera margins on Venus we describe here, or the presence or nature of any detachment beneath the narrow folds—although were oceans ever present on Venus, their loss might have left vast evaporite deposits behind (Wood & Amsbury, 1986). Nevertheless, the general trend in folds widening from plains to tessera at each of the sites in Figures 2–5 accords with greater depths of faulting from thin- to thick-skinned tectonics with increasing proximity to the tessera proper. This inference agrees with the proposal by Suppe and Connors (1992) that the brittle lithosphere thickness and depth extent of faulting increase with greater elevation and, presumably, crustal thickness, on the basis of a strong correlation between topography and the relief of select shortening systems assessed with critical-taper-wedge mechanics. By this hypothesis, the pattern of progressive shortening from the plains to the tesserae described here is consistent with a single deformation mechanism. Moreover, the fact that there are relatively narrow folds interspersed with the larger anticlines (see: Figure 2, panels e and f; Figure 3, panel d; Figure 4, panel d; and Figure 5, panel c) probably reflects the substantial structural complexity inherent to fold-and-thrust belts (e.g., Poblet & Lisle, 2011; Suppe & Connors, 1992) (Figure 6, panels d–f), such that the anticlines here are not only the result of simple buckle folding (cf. Biot et al., 1961).

By analogy with these and other continental sites of crustal shortening on Earth, then, and on the basis of folds becoming broader but retaining the same general trend toward the highlands, we interpret the tessera margins at Nayunuwi Montes and Uorsar Rupes and along northern Thetis Regio and the Sogolon Planitia–Aphrodite Terra boundary as comprising fold-and-thrust belts that show progressively greater fault penetration depths from the plains to the tesserae, a conclusion similar to but reached by independent means from that of Suppe and Connors (1992).

We thus hypothesize that, along portions of the margins of the two largest highland terrae on Venus, increases in the area of tessera terrain and associated uplift and thickening of the underlying crust have taken place by the shortening and incorporation of stratigraphically young smooth plains. The fact that some of the smallest resolvable folds seem to postdate the plains units indicates that a portion of this deformation has taken place in the geologically recent past—that is, at times following the emplacement of those incorporated plains materials.

Our view here is different from earlier analyses of the margins of Ovda Regio. For example, Romeo and Capote (2011) concluded that shortening had essentially ended prior to the emplacement of the locally youngest smooth plains. Hansen (2006) attributed folds of different wavelengths in the tesserae to the shortening of a progressively thickening layer of solidified impact melt. Ivanov and Head (1996) noted the post-emplacement tilting of plains at this and some other tessera boundaries, but offered no kinematic interpretation of this deformation. Moreover, whereas Ivanov and Head (2011) assigned plains, ridges, and tessera terrain blocks to different time-stratigraphic systems, we instead suggest that these units differ mainly in terms of the amount of shortening strain to which they have been subjected but could otherwise be similar in emplacement age and lithology, at least at the four sites above.

4.2. Implications for Tessera Formation and Composition

An enduring question regarding tesserae has been the nature of their precursor rocks (e.g., Ivanov & Head, 1996). Per our interpretation, at least some tessera units are highly deformed remnants of former radar-smooth plains units, which themselves were likely derived from effusively emplaced lavas and/or igneous-derived sediments (e.g., Carter et al., 2023; Ivanov & Head, 2011). The concept of volcanic plains transforming to tessera provides a possible explanation for observations of layering within some tesserae, that is, that those layers may be the component flows or flow units that originally formed smooth plains (Byrne, Ghail, Gilmore, et al., 2021). Indeed, in addition to fold wavelengths varying in response to the geometrically complex thrust faulting that characterizes fold-and-thrust belts on Earth (Figure 6, panels d–f), the shortening of layered plains units could also help account for the observation of folds of multiple wavelengths within tesserae (e.g., Ghent & Hansen, 1999; Romeo & Capote, 2011), with those wavelengths reflecting different thicknesses of folded packages of lavas controlled, in turn, by laterally varying strengths and depths of mechanical discontinuities such as lava interbeds. Salts, shales, and other mechanically incompetent strata act as décollements in shortening systems on Earth; as we note, however, the mechanical structure of the Venus subsurface at tessera margins and the nature of any detachment surfaces there remain unknown.

We can document only horizontal shortening rather than the motions of plains or highlands within any mantle reference frame. Nevertheless, the extensive region of recent shortening along the northern Aphrodite margin contrasts sharply with—and may serve to balance, at least in part—the recent horizontal extension that has occurred across the immense rift zones that constitute the southeastern region of that highland. Further, there is structural evidence that portions of Ovda Regio and the Tellus Regio tessera were assembled from smaller, antecedent pieces of formerly separated tessera terrain (Chadwick & Schaber, 1994; Gilmore & Head, 2018), and smooth plains appear to have been thrust onto the southeast margin of Thetis Regio, in eastern Aphrodite Terra (Ghail, 2002). Large-scale horizontal mobility of terrain on Venus, including portions of present tesserae, may at one time have been more widespread than has generally been assumed in models for the geological evolution of Venus.

Importantly, not all tessera boundaries show this type of deformational boundary. Ivanov and Head (1996) determined that about three-quarters of tessera boundaries appear to have been embayed by younger plains without noticeable subsequent deformation. Other plains–tessera boundaries display different tectonic histories from those documented here. For instance, the western boundary of Alpha Regio features a margin-parallel shortening fabric (Gilmore & Head, 2000), but there is no spatially contiguous continuation of that fabric into the surrounding plains units (Figure 1, panel a) as we document for Nayunuwi Montes, Uorsar Rupes, northern Thetis Regio, and Sogolon Planitia.

Moreover, geologically recent tessera formation does not mean that all portions of the associated exposures of tesserae are young. In fact, the central regions of those units may well be quite old, as suggested by the average areal densities of large impact craters in the tessera terrain (Ivanov & Basilevsky, 1993). The areal density of smaller impact craters ($D < 16$ km), however, is lower on tesserae than on plains, and some of the larger ($D = 40$ – 105 km) craters in tesserae show evidence of having been tectonically deformed (Ivanov & Basilevsky, 1993). Earlier workers (Ivanov & Basilevsky, 1993; Schaber et al., 1992) ascribed the primary cause of the deficiency of smaller craters ($D < 16$ km) within tesserae to a bias caused by the difficulty in recognizing the radar-bright ejecta deposits of such features on this radar-bright terrain type. Alternatively, smaller craters formed in rocks that were later transformed to tessera may have been rendered unrecognizable by that deformation, as well as by the erosion implied by the exposure of constituent layering within tesserae (Byrne, Ghail, Gilmore, et al., 2021). This reasoning supports the view that deformation continued in some highland tesserae for an extended period, including at least a portion of the time following local plains emplacement.

5. Outlook

By attributing several examples of seemingly morphologically and temporally disparate units on Venus—plains, ridges, and tesserae, which had earlier been assigned to different periods in the planet's geological history (e.g., Ivanov & Head, 2011) (Figure 7)—to discrete kinematic stages of a common process, we propose that at least some tessera exposures may reflect long-term, sustained crustal shortening that involves rocks of different lithologies, emplacement mechanisms, ages, and/or weathering histories, rendering inappropriate any global-scale correlation of mapped units solely on the basis of structural characteristics (Figure 7, panels b–e). Such a

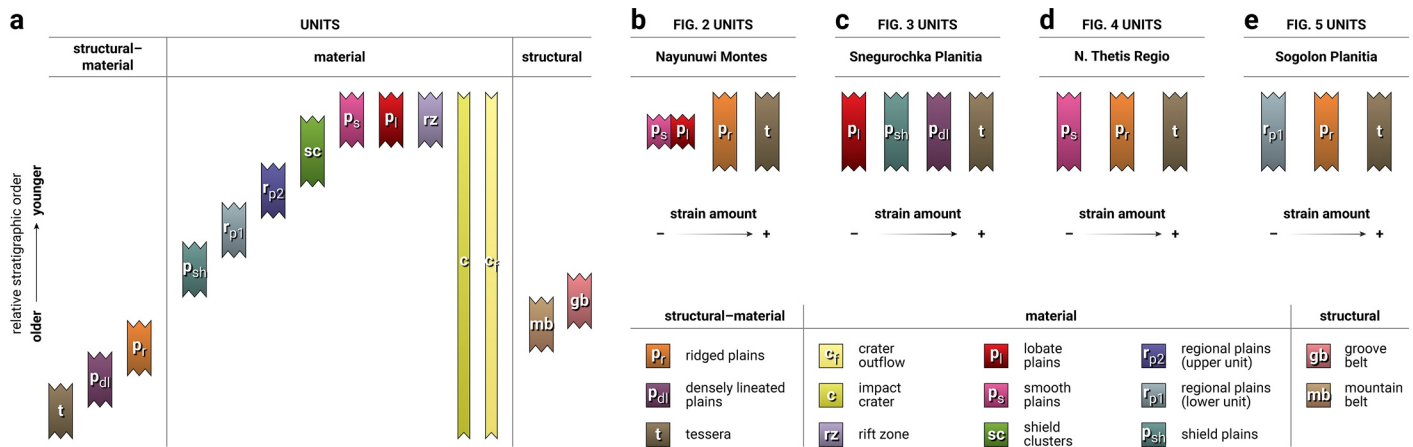


Figure 7. Comparison of two interpretations of mapped units along the plains–tessera margins of this study. (a) The relative stratigraphic order of global units mapped by Ivanov and Head (2011) (modified from that reference). The units are arranged from oldest (bottom) to youngest (top). (b) Those mapped units at the region of tessera formation along Nayunuwi Montes (Figure 2), interpreted here as featuring approximately contemporaneous deformation and ordered left to right from least to greatest strain accommodated by mapped structures. Note that we regard the lobate and smooth plains units here as having been approximately equally strained. (c–e) Those mapped units at the regions of tessera formation shown in Figures 3–5, at Snegurochka Planitia, Thetis Regio, and Sogolon Planitia, respectively, again ordered from least to greatest strain.

scenario is consistent with observed differences in radiophysical—and so possibly mineralogical—properties of different tesserae across Venus (Brossier & Gilmore, 2021). If so, then prevailing ideas for the global geological evolution of Venus (Basilevsky & Head, 1998; Ivanov & Head, 2011), levels of recent geological activity on the planet (Smrekar et al., 2018), the formational history of the tesserae, and locations within these exposures to which we should look for any preserved geochemical or morphological evidence of an earlier era with an Earth-like climate (Hashimoto et al., 2008; Way & Del Genio, 2020; Way et al., 2016), should be revisited.

Early in the next decade, NASA's Venus Emissivity, Radio Science, InSAR, Topography, and Spectroscopy (Smrekar et al., 2022) spacecraft and the European Space Agency's EnVision (Ghail et al., 2018) mission are slated to acquire orbital radar image and topographic data of the Venus surface at resolutions markedly superior to those of Magellan. EnVision will also carry a radar sounder (Bruzzone et al., 2020) to characterize the structure of the upper few hundred meters of the Venus subsurface. Detailed mapping of superposition relations with high-resolution radar imagery will help to determine the extent to which stratigraphically young plains have been folded. Measurements of strain with high-resolution topography along transects parallel to the dominant shortening axes at plains–tessera boundaries will enable quantitative comparisons between these systems and fold-and-thrust belts on Earth. And searching for and characterizing how subsurface interfaces vary geometrically with distance from these boundaries will inform numerical models of crustal shortening on Venus (e.g., Crane & Byrne, 2025) and could reveal whether the transition we propose from thin- to thick-skinned tectonics from the plains to the tessera is present. Collectively, these new data sets will provide a critical basis for testing our hypothesis of geologically recent formation of some tessera terrain on Venus by the tectonic processing of volcanic plains.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

All data derived in this study are included in the main text or described in the methods. The Magellan synthetic-aperture radar (SAR) east- and west-looking global radar mosaics are available at https://astrogeology.usgs.gov/search/map/venus_magellan_sar_fmap_left_look_global_mosaic_75m and https://astrogeology.usgs.gov/search/map/venus_magellan_sar_fmap_right_look_global_mosaic_75m, respectively.

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